Bespoke Model Development

Sample details:
The model was developed on 75% and validated on 25% of the sample.

Observation Period	July 2023 – March 2024
Development (75%)	70,698
Validation (25%)	23,566

Good/Bad Definition:

GB Flag Value	**GB Flag Definition**
Outcome Period	1 month
Bad	DPD30+ - Clients who delayed instalment with more than 30 days after first month
Good	DPD30<30 - Clients who have not delayed instalment with more than 30 days after first month

Development Sample	
Goods	60,575
Bads	10,123
Sample Bad Rate	14.3%
Validation Sample	
Goods	20,236
Bads	3,330
Sample Bad Rate	14.1%

Bespoke model - Score Distribution -Train:

Gini: 36.8%

Band	Score Range	Bad	Good	All	Bad Rate
1	300-502	2418	4553	6971	34.7%
2	503-521	1518	5463	6981	21.7%
3	522-532	1259	5700	6959	18.1%
4	533-540	1079	5825	6904	15.6%
5	541-546	898	5763	6661	13.5%
6	547-552	826	6455	7281	11.3%
7	553-558	744	6849	7593	9.8%
8	559-563	529	5556	6085	8.7%
9	564-570	485	6915	7400	6.6%
10	571-600	367	7496	7863	4.7%

Bespoke model - Score Distribution – Test:

Gini: 30.6%

Band	Score Range	Bad	Good	All	Bad Rate
1	300-502	723	1597	2320	31.2%
2	503-521	489	1883	2372	20.6%
3	522-532	371	1917	2288	16.2%
4	533-540	338	1935	2273	14.9%
5	541-546	278	1862	2140	13.0%
6	547-552	295	2106	2401	12.3%
7	553-558	283	2275	2558	11.1%
8	559-563	211	1920	2131	9.9%
9	564-570	197	2271	2468	8.0%
10	571-600	145	2470	2615	5.5%

Bespoke model - Score Distribution - Train + Test:

Gini: 34.6%

Band	Score Range	Bad	Good	All	Bad Rate
1	300-502	3141	6150	9291	33.8%
2	503-521	2007	7346	9353	21.5%
3	522-532	1630	7617	9247	17.6%
4	533-540	1417	7760	9177	15.4%
5	541-546	1176	7625	8801	13.4%
6	547-552	1121	8561	9682	11.6%
7	553-558	1027	9124	10151	10.1%
8	559-563	740	7476	8216	9.0%
9	564-570	682	9186	9868	6.9%
10	571-600	512	9966	10478	4.9%

Post-paid customers

Score Band	Score	Risk Level
21	880	High
22	920	Medium
23	960	Low

Model Development Stages:

- **Data audit and preparation**
  Cleaning the Telco data from data with lower quality (not populated records, etc.). Deriving a set of features based on the raw Telco data.
- **Target definition**
  Defining the target definition. This step requires collaboration with the client's Risk team
- **Sample selection**
  Choosing the most appropriate sample window based on the Target definition, seasonality, past marketing campaigns, etc.
- **Features selection**
  Creating a short list with features to be tested in the model development
- **Model Training on 75%**
  Building a model using only 75% of the Accepted accounts – Probability of Default
- **Model Validation on 25%**
  Validating the model using the remaining 25% of the Accepted accounts – Probability of Default

Machine learning techniques used in Model Development:

- FinScore initially develops a Logistic Regression model and use its predictive power as a base. Then attempts various ML algorithms until coming up with the best possible result
- ML techniques that are most frequently attempted:
  - XGBoost
  - Random Forrest (usually when the sample size it small)
  - RBM Neural Network
- ***Current ML technique that outperformed the rest and was finally selected was XGBoost***

Telco Data groups:

FinScore receives more than 400 raw features from Smart for every single cellphone number. Additional ~200 features are derived from the raw data. The following groups of Telco data are included:

- **Voice usage** – outgoing calls (minutes, number of calls, duration, etc.)
- **SMS usage** - outgoing SMSs (number, time of the day, etc.)
- **Mobile Data usage** – used mobile data (megabytes, frequency, etc.)
- **Top-up patterns** – recharges (amounts, number of recharges, frequency, etc.)
- **Sim card data** – activation date, last usage, etc.)
- **Behavior patterns** – sending load to other individuals, topping up from different sources, etc.)
- **Spending patterns** – amounts, packages/bundles, frequencies, etc.)
- **Location data** – daytime, nighttime, weekend location
- **Device data** – device main parameters – smartphone/feature phone, screen size, CPU, etc.
- **Social circle** – The frequently contacted numbers via voice, SMS and Pasa Load (the service of sending load/airtime to other individuals